CUNI at AlexandriaX-2026: Finetuning LLMs for dialectal Arabic Translation

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Abstract

We describe the Charles University (CUNI) submission to the AlexandriaX-2026 Subtask 1, a shared task on English-to-Arabic dialectal conversation translation covering 13 Arabic varieties. Our approach fine-tunes Gemma-4-31B-it with QLoRA on the Alexandria training data at the conversation level, using rich contextual prompts that include domain metadata, participant names and speaker labels. We explore multiple training data mixes and hyperparameters, yielding a pool of candidate translations for the test set. These are reranked with a genetic-algorithm-enhanced minimum Bayes risk (GA-MBR) decoding using sentence-level spBLEU as the fitness function. Fine-tuning improves over the unmodified base model by approximately 5.2 ChrF points on the development set (from 45.2 to 50.4 ChrF for the best single checkpoint, or 25.1 to 32.6 spBLEU); GA-MBR reranking over a 40-candidate development pool yields 51.3 ChrF / 34.0 spBLEU, a further improvement of 0.9 ChrF points over the best pool member. Our submission ranked second in BLEU/chRF++ the unconstrained track.

1 Introduction

The AlexandriaX-2026 Challenge Subtask 1 (El Mekki et al., 2026) is a shared task on translating English conversations into 13 Arabic dialectal varieties (listed in Table 1). The task uses the UBC-NLP/Alexandria dataset (EL Mekki et al., 2026), which provides multi-turn bilingual conversations in English and one of these varieties.

There are two aspects of the task that should be kept in mind by the participants. First, translations must preserve conversational coherence across turns: a good translation of one turn may depend on what was said in previous turns, and also on the people involved in the conversation, their relationship, and the domain and topic. All of this additional information is part of the dataset. Second, dialectal appropriateness is important — the same English sentence should be rendered differently depending on the target variety.

Our submission builds on QLoRA fine-tuning and MBR reranking in combination with a genetic algorithm to modify the sentences in order to improve automatic scores. The main evaluation metric of the task is spBLEU with a single, human-made reference. This makes the conclusions based on the final ranking weaker (compared to human evaluation), as differences in the score might not reflect an actual translation quality change.

The paper is organized as follows. Section 2 describes the task, dataset, and related work. Section 3 presents our system. Section 4 covers the experimental setup. Section 5 reports results.

2 Background

2.1 Task Setup

The AlexandriaX-2026 Subtask 1 is a shared task on translating English conversations into 13 Arabic dialectal varieties.

Evaluation. The evaluation metrics are spBLEU (SentencePiece BLEU with the flores200 model; Goyal et al. 2022; Papineni et al. 2002; Team et al. 2022) and character n-gram F-score. ChrF, i.e., word n-gram order equal to 0 is used for our experiments; ChrF++, i.e. word n-gram order equal to 2, is used in the official evaluation (Popović, 2015).

Tracks. The challenge offers constrained and unconstrained tracks distinguished by allowed training data and model size. Our submission participates in the **unconstrained** track: we fine-tune Gemma-4-31B-it (Team, 2026), which exceeds the 5B parameter limit of the constrained track.

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2.2 Datasets

Alexandria. The Alexandria dataset (EL Mekki et al., 2026) consists of multi-turn conversations in English, generated by an LLM, paired with their dialectal Arabic translation created by humans, across 13 target varieties. Conversations are short (2–4 turns) and annotated with a domain label (e.g. *business*, *technology*, *healthcare*) and participant names. The dataset is split into training, development, and public/private test portions; Table 1 summarises the sizes. We use the development split (3,963 conversations, 12,250 turns) for all reported evaluations.

WMT23 news data. We created a synthetic dialectal Arabic dataset by translating the English source sides of WMT23 test sets (Kocmi et al., 2023). We translated each English sentence into all 11 task dialects using Gemma-4-{12,31}B-it, Jais-2-70B-Chat and gpt-5.4-nano, then scored every translation with a GEMBA-style judge (Kocmi and Federmann, 2023) (Gemma-4-31B-it rating translation accuracy and dialect authenticity on a 0–100 scale) and retained only the best translation for each source sentence, and only if the translation scored ≥ 90 on both dimensions. Because the same model (Gemma-4-31B-it) acts as one of the translators and the judge, the filter may over-select outputs produced by this model.

Synthetic conversations. We generated 43,978 synthetic bilingual conversations using GPT-5.4-mini, seeded from Alexandria metadata (domain labels, participant roles, turn counts). These synthetic conversations cover also the Libyan and Sudanese dialects, that are excluded from the original Alexandria training dataset.

	Train		Dev		Public Test	
Dialect	**Conv**	**Turns**	**Conv**	**Turns**	**Conv**	**Turns**
Egyptian (EG)	982	3,108	352	1,113	366	1,118
Jordanian (JO)	1,730	5,501	352	1,113	347	1,107
Lebanese (LB)	2,915	8,906	371	1,118	369	1,106
Libyan (LY)	—	—	—	—	374	1,109
Moroccan (MA)	815	2,573	354	1,110	353	1,115
Mauritanian (MR)	1,748	5,515	352	1,114	357	1,112
Omani (OM)	2,066	6,280	381	1,109	387	1,118
Palestinian (PS)	4,669	14,933	352	1,110	370	1,109
Saudi (SA)	2,699	8,470	363	1,110	351	1,113
Sudanese (SD)	—	—	—	—	345	1,106
Syrian (SY)	1,869	6,071	347	1,119	353	1,114
Tunisian (TN)	665	2,034	378	1,116	379	1,109
Yemeni (YE)	988	3,089	361	1,118	355	1,106
Total	**21,146**	**66,480**	**3,963**	**12,250**	**4,706**	**14,442**

Table 1: Alexandria dataset split sizes by dialect. Libyan and Sudanese Arabic appear only in the test set.

2.3 Related Work

Arabic dialectal machine translation. Translating into Arabic dialects has received growing attention alongside the development of dialectal corpora such as MADAR (Bouamor et al., 2018). Most prior work focuses on dialect identification (Salameh et al., 2018; Abdelali et al., 2021) or translation *from* dialects into MSA or English (Zbib et al., 2012; Sajjad et al., 2013), rather than translation *into* dialects. The Alexandria Challenge targets the harder direction of producing dialectally appropriate output from English. More recently, large language models finetuned for Arabic have pushed quality further (Nagoudi et al., 2022a,b; Anwar et al., 2026; Jon et al., 2026).

Minimum Bayes risk decoding. Minimum Bayes risk (MBR) decoding selects the hypothesis that maximises expected utility under a distribution approximated by a set of samples (Kumar and Byrne, 2004). In case of reference-based metric used as utility, it selects the hypothesis that scores the best using all the other hypotheses as the references. Applied to MT, MBR consistently outperforms MAP (beam search) decoding (Eikema and Aziz, 2020; Freitag et al., 2022). An extension of this idea is to use a genetic algorithm to increase the number of high-quality candidate translations (Jon and Bojar, 2023).

3 System overview

3.1 Model Selection

We evaluated three Arabic-capable instruction-tuned models as base systems without fine-tuning (*Base* columns in Table 2): Gemma-4-31B-it and Gemma-4-12B-it (Team, 2026), and Jais-2-8B-Chat (Anwar et al., 2026). Gemma-4-31B-it and Gemma-4-12B-it score similarly out of the box (45.2 and 45.4 ChrF respectively), while Jais-2-8B-Chat lags at 33.1 ChrF, with near-zero score on Mauritanian Arabic (2.4 ChrF). We selected Gemma-4-31B-it as our primary model.

3.2 Conversation-level Translation

We translate each conversation as a single unit rather than turn by turn. The model receives all English turns in one prompt and outputs all Arabic turns in one pass, giving it access to full discourse context.

Rich context prompting. We prepend a context header to each conversation and label every turn,

based on the additional info provided in the input:

```
You are a professional, very precise translator
and a native speaker. Translate inputs based
on the instructions and always print out only
the text of the best possible translation, no
explanations. Keep the same formatting (e.g.
markup, lines, spacing) as the original. Do
not translate untranslatable parts of the input
(URLs, code and similar).
Translate the following text into
{dialect} dialect, only print out the
translation, do not add any explanations:
[Domain:{domain}|Participants:{participants}]
{1st_speaker}: {turn_1}
{2nd_speaker}: {turn_2}
. . .
```

3.3 Fine-tuning

We fine-tuned the models with (Q)LoRA (Dettmers et al., 2023) with multiple hyperparameter configs: (LoRA rank $r \in \{8, 32, 64, 128\}$, α=$2r$, full bfloat16 or 4-bit quantisation, $lr \in \{$2e-5, 6.36e-5, 1e-4$\}$). Training used effective batch size 64 (4×16 gradient accumulation steps), maximum sequence length 768 tokens, and up to 8,000 steps with checkpoints evaluated every 150 steps on a 150 dev set conversations per dialect. All 11 dialects were included simultaneously.

Training data. We experimented with five dataset mixes: `alx` (Alexandria training split only); `alx+wmt23` (Alexandria augmented with GEMBA-filtered WMT23 dialectal translations); `alx+synth` (Alexandria augmented with the synthetic GPT-5.4-mini conversations); `alx+wmt23` (Alexandria augmented with GEMBA-filtered WMT23 dialectal translations and the MADAR dataset) and `alx+test` (Alexandria train + public test set, to see if more in-domain data improve the scores, was not used in the final submission).

3.4 GA-MBR Ensemble

We further increased the evaluation scores by using genetic-algorithm minimum Bayes risk decoding (GA-MBR; Jon and Bojar 2023). We constructed a pool of candidate translations from multiple checkpoints and data configurations and combined them to obtain better MBR scores.

Each candidate is treated as a GA gene. The fitness function is sentence-level spBLEU (Goyal et al., 2022), computed against all remaining candidates as pseudo-references. The candidate translations are mutated by insertion, deletion or replacement operations and crossed-over with each other over multiple generations.

4 Experimental Setup

Preprocessing. Conversations are formatted as described in Section 3, loss is computed only on the Arabic output tokens.

Software and hardware. Training uses `transformers` (Wolf et al., 2020), `PEFT` (Mangrulkar et al., 2022), and `TRL` (von Werra et al., 2020) on a single NVIDIA GPU (either A100, A40 or H100, depending on the node). Translations are generated with `vLLM` (Kwon et al., 2023). 4-bit QLoRA quantisation used `bitsandbytes` (Dettmers et al., 2023). Evaluation uses `sacrebleu` (Post, 2018).

Decoding and post-processing. At inference time we use greedy decoding with thinking mode disabled (`enable_thinking=False`). The model is prompted to produce output in the same numbered-turn format as the training target.

GA-MBR As the search for optimal hyperparameters for the GA would be too resource-consuming, we set the mutation rate to 2 (2 tokens of each candidate on average are mutated) and crossover probability to 0.2 for all the experiments. The GA runs for 200 generations for the dev set experiments and 140 generations for the final submission. The GA operates independently at the *turn* level: each source turn is processed as a separate instance with all candidate translations of that turn as the initial population and pseudo-references (each initial candidate is copied 20 times to increase the population size, resulting in population sizes of 800 and 960 for our experiments). Candidates are tokenized into flores200 SentencePiece subword pieces (the same tokenizer used by spBLEU), which form the genes for crossover and mutation. Crossover performs single-point splicing of two parent gene sequences; mutation substitutes individual subword tokens drawn from the union of all candidates' subword vocabularies. No semantic, fluency, or adequacy constraint beyond spBLEU fitness is applied. The final output for each turn is the best-scoring evolved sentence after all generations; this sentence may be a novel combination of subwords not present verbatim in any of the original candidates. Turns across a conversation are assembled independently, so different turns may originate from different candidates or from evolved hybrids.

We also experimented with conversation-level processing for the GA, where we concatenated the

whole conversation and treated it as a single gene for scoring and mutation/crossover, but this approach yielded worse final scores in our preliminary experiments.

The 48 test-set candidates came from multiple checkpoints of Gemma-4-31B-it fine-tuned on `alx`, `alx+synth`, `alx+wmt23` or `alx+wmt23+madar` at various learning rates and steps, selected based on dev set chrF scores, see Table 6 in Appendix B. The development-set ablation pool contained 40 candidates selected by the same key.

Evaluation metrics. We report **ChrF** (Popović, 2015) and **spBLEU** (corpus-level BLEU with SentencePiece flores200 tokenisation).

5 Results

Base model performance. Gemma-4-12B-it and Gemma-4-31B-it score nearly identically without fine-tuning (45.4 vs. 45.2 ChrF overall). Jais-2-8B-Chat scores substantially lower (33.1 ChrF), with particularly low scores on Mauritanian (2.4 ChrF) and Omani (20.6). These low ChrF values show the model is producing outputs far from the references for these varieties, possibly due to limited exposure to them in pre-training.

Effect of fine-tuning. QLoRA fine-tuning yields consistent ChrF gains across all three models. Jais-2-8B-Chat shows the largest score increase overall (33.1 → 46.7 ChrF, +13.6), with especially large gains on Mauritanian (+30.3) and Omani (+28.1) — varieties where the base model produced near-zero scores. Gemma-4-31B-it gains 5.2 ChrF points (45.2 → 50.4), while Gemma-4-12B-it gains 3.9 points (45.4 → 49.3). The fine-tuned Gemma-4-31B-it reaches the highest ChrF overall (50.4 / 32.6 spBLEU), though the margin over Gemma-4-12B-it is small (1.1 ChrF), which may suggest that data quality and quantity have a stronger effect than model size in this range.

Other datasets. None of the additional datasets led to an increase in dev set spBLEU scores over the best single checkpoint trained only on the Alexandria dataset, the detailed results are shown in Table 5 in Appendix A. Some of the checkpoints trained either on `alx+synth` or `alx+wmt23` configuration were however part of the top 48 candidates in the final test set pool for the GA-MBR.

Cross-dialect variation. Syrian Arabic consistently receives the highest scores across all fine-tuned systems (53–56 ChrF), while Mauritanian Arabic scores lowest (30–38 ChrF). Whether this reflects low translation, low reference quality, or distribution mismatch between the training data and these varieties cannot be determined from automatic scores.

GA-MBR. Table 3 shows the effect of GA-MBR reranking over a 40-candidate development pool. MBR (equivalent to GA-MBR with g=0) already yields a higher score than the highest-scoring individual pool member (33.9 vs. 32.6 spBLEU). Evolutionary optimisation for 200 generations adds a marginal further increase (34.0 spBLEU), suggesting that score improvement from the GA step is small relative to MBR pool diversity.

Test set results. On the private test set (Table 4), the best single checkpoint scores 31.42 spBLEU. MBR reranking over a 48-candidate pool raises this to 32.47, and GA-MBR optimisation for 140 generations adds a further 0.05 points (32.52). The candidate checkpoint were selected based on their dev set ChrF scores, the list of the checkpoints is shown in Appendix B, Table 6.

6 Conclusion

We presented the CUNI submission to the AlexandriaX-2026 Subtask 1 on English-to-dialectal Arabic conversation translation. Our system combines QLoRA fine-tuning of Gemma-4-31B-it in conversation-level translation mode with a GA-MBR over a pool of candidate translations.

Fine-tuning with Alexandria dataset yields a 5.2 ChrF score increase over the base model (45.2 → 50.4). GA-MBR reranking over 40 development candidates adds a further 0.9 ChrF (51.3), with the gain largely attributable to MBR pool diversity rather than the evolutionary search itself. Mauritanian Arabic consistently receives the lowest automatic scores. Our final submission was ranked second based on spBLEU.

Limitations

Our evaluation relies primarily on automatic reference-based metrics, with only a single human reference per source, and does not include human evaluation of adequacy, fluency, or dialectal authenticity. Consequently, improvements in spBLEU or ChrF may not always correspond to improvements in translation quality.

Dialect	G4-12b				G4-31b				Jais-2-8b-chat			
	Base		*SFT*		*Base*		*SFT*		*Base*		*SFT*	
	ChrF	**spBLEU**	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**
EG	50.7	31.8	53.2	36.9	50.6	31.6	53.4	37.4	43.1	25.8	50.4	34.0
JO	50.4	29.7	54.0	37.1	50.1	29.3	55.5	38.5	39.8	20.4	51.8	34.5
LB	46.6	26.8	49.7	32.9	47.1	26.7	50.0	33.4	35.5	18.3	46.8	29.3
MA	40.8	21.1	43.7	25.3	40.8	21.1	45.1	26.9	30.8	14.5	41.7	23.5
MR	30.6	9.4	37.2	16.6	30.5	9.1	37.9	15.9	2.4	0.5	32.7	10.6
OM	43.2	21.8	49.2	28.6	41.9	20.4	53.8	35.9	20.6	7.1	48.7	29.5
PS	49.5	29.6	52.4	34.8	49.0	29.1	53.3	36.0	36.0	18.4	50.3	32.1
SA	50.7	29.5	54.4	36.0	49.7	28.6	55.7	37.7	44.5	23.7	51.4	32.5
SY	52.7	32.1	55.4	38.3	51.2	30.7	55.6	38.8	48.2	28.6	52.9	35.5
TN	40.9	22.1	47.4	29.9	42.8	24.6	48.2	30.8	35.1	16.7	44.9	26.6
YE	44.7	23.9	46.4	27.3	44.6	23.8	47.6	28.7	31.2	13.1	44.6	25.0
ALL	**45.4**	**25.3**	**49.3**	**31.3**	**45.2**	**25.1**	**50.4**	**32.6**	**33.1**	**16.4**	**46.7**	**28.1**

Table 2: Per-dialect ChrF and spBLEU on the Alexandria development set for both the base models and the best finetuned checkpoints for each base model (selected by an overall chrF on a subset of the dev set, per-dialect best scores might be higher).

System	ChrF	spBLEU
Base G4-31B-it	45.20	25.07
SFT G4-31B-it	50.42	32.61
MBR g=0 (40 dev candidates)	51.18	33.87
GA-MBR g=200 (40 dev candidates)	**51.26**	**33.96**

Table 3: Development-set ablation (11 dialects, 12,250 turns). The pool contains 40 candidates from multiple checkpoints and data configurations, none trained on the test set. GA-MBR g=0 is MBR initialisation without evolutionary optimisation.

System	spBLEU
Best single checkpoint	31.42
MBR (g=0, 48 candidates)	32.47
GA-MBR* ($g \sim 100$, 48 candidates)	**32.5**
GA-MBR (g=140, 48 candidates)	**32.52**

Table 4: Private test set spBLEU scores. The candidate pool contains 48 translations from the best checkpoints across `alx`, `alx+synth`, `alx+wmt23` and `alx+wmt23+madar` data configurations. Asterisk shows the final submission, where the GA process has not fully finished and ran for around 100 generations for most of the examples.

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A Different dataset mixes

Table 5 compares the four training data configurations for Gemma-4-31B-it. The overall scores are close: `alx`, `alx+wmt23`, and `alx+synth` all fall within 0.3 ChrF of each other (50.4, 50.2, and 50.2 respectively), with no configuration consistently dominating at the dialect level. `alx` achieves the highest aggregate ChrF and performs best on Omani, Saudi, and Palestinian varieties; `alx+wmt23` leads on Egyptian, Syrian, and Tunisian; and `alx+synth` scores highest on Lebanese. Adding the public test set (`alx+test`) reduces development set performance by about 0.6 ChrF.

B Final candidate pool

Table 6 shows the composition of the initial list of translations (checkpoints used to produce these translations) used for the final GA-MBR for the submission.

	alx		alx+wmt23		alx+synth		alx+test	
Dialect	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**	**ChrF**	**spBLEU**
EG	53.4	37.4	**53.8**	**37.7**	53.4	37.4	53.7	37.5
JO	**55.5**	38.5	55.2	38.3	55.0	38.2	55.4	**38.9**
LB	50.0	33.4	49.4	32.3	**50.9**	**34.5**	50.0	33.0
MA	45.1	26.9	45.2	26.8	44.8	26.5	**45.6**	**27.3**
MR	**37.9**	15.9	37.5	16.2	37.5	**16.5**	36.7	15.6
OM	**53.8**	**35.9**	51.3	32.1	51.5	32.5	49.0	28.1
PS	**53.3**	**36.0**	52.8	35.1	53.1	35.7	53.2	35.4
SA	**55.7**	**37.7**	54.8	36.7	55.0	36.9	54.5	35.8
SY	55.6	38.8	**56.4**	**38.9**	55.5	38.4	55.6	38.2
TN	48.2	30.8	**49.4**	**32.3**	48.6	32.0	48.6	31.3
YE	**47.6**	**28.7**	47.5	28.4	**47.6**	**28.7**	46.8	27.7
ALL	**50.4**	**32.6**	50.2	32.3	50.2	**32.6**	49.8	31.7

Table 5: Per-dialect ChrF and spBLEU on the Alexandria development set (11 dialects, LY and SD excluded) for Gemma-4-31B-it fine-tuned on four training data configurations: `alx` (Alexandria training split only), `alx+wmt23` (augmented with GEMBA-filtered WMT23 dialectal translations), `alx+synth` (augmented with GPT-5.4-mini synthetic conversations), and `alx+test` (Alexandria train + public test set, included for completeness — see text). Each column shows the best checkpoint per configuration. Bold marks the highest value in each row.

Data mix	**LR**	**Steps**
`alx`	5e-5	450, 600, 600, 600, 600, 750, 900, 900, 1050, 1050, 1200, 1350, 1500
	8e-5	250, 600
`alx+synth`	8e-5	300, 300, 350, 900
	9e-5	300, 450
`alx+wmt23`	5e-5	1650, 2100, 2400, 3000, 3900, 4200, 5400, 5550, 5700, 5850, 6000, 6150, 6300, 6450, 6600, 6750, 6900, 7050, 7200, 7350, 7500, 7650, 7800, 7950
	8e-5	1650
`alx+wmt23+madar`	8e-5	1500, 1650

Table 6: All 48 candidates in the test-set GA-MBR pool (Gemma-4-31B-it). `+synth` adds GPT-4-mini synthetic conversations; `+madar` adds the MADAR corpus. Multiple entries with the same LR come from different training runs. All candidates use $r = 128$.